Sreeram_experimental
Branch Overview & Model Workflow

Concrete description of how raw OHLCV + signals + Bloomberg data become a locked, out-of-sample-validated meta-model and a deliverable strategy CSV for the 11 instruments in the universe — plus the data, features, models, importance, and performance tables for the submission-readiness review.

Contents

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1 What the model does

The meta-model is the **secondary layer** in a Lopez de Prado meta-labelling setup. A primary signal (in `primary_signals.csv`) already decides **direction** for every day — short, flat, or long. Our model does **not** predict direction or return. For every event (each day the primary signal is non-zero) it answers one binary question: **given that the primary signal wants to take this trade, is it worth taking?**

- **Label** comes from the triple barrier (López de Prado AFML Ch. 3): from the event day we apply an upper (profit), lower (stop), and vertical (time) barrier in the signal's direction. The label is **1 if the trade hits its profit barrier first, 0 otherwise**.
- **Features** are the full F1–F22 + EWMA-HMM stack (80 features after a drift filter), computed causally up to each event's timestamp.
- **Rows** are the events (non-zero-signal days). After dropping events whose held window is not fully observable, we have **4,886 events** across 11 instruments (63–626 per instrument).
- **Score** is AUC — the model's ability to rank profitable signals above unprofitable ones. AUC ≈ 0.5 means no filtering skill; a lower 1-SE CI above 0.5 is what we call **signal**.

The model is a **precision filter on an existing strategy**, not a return predictor.

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2 Guiding principle: train-only selection, single-shot out-of-sample

Every fitted object — feature drift filter, sigma estimator, hyperparameter grid, model family, calibrator, position sizer — is fit on **train data only**. The held-out test set is **sealed**: read once, at the end, to produce the deliverable. Selecting among variants on the test set would re-contaminate it (winner's curse), so all selection happens on train via **CombinatorialPurgedCV(6, 2)** $\rightarrow$ 15 paths, and the locked choice per instrument is committed **before** the sealed slice is touched.

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3 The pipeline

3.1 Stage 1 — Train / test split

A single **global chronological cut at** `2021-10-06` (70th percentile of the 4,886 pooled events), the same for every instrument so that individual-vs-pooled and cross-instrument comparisons share one test window.

- **Purge** — drop train events whose label horizon `t_end` crosses past the cut. Removes look-ahead from the barrier horizon.
- **Embargo** — 10 trading days after the cut (`2021-10-20`). The post-embargo 30% slice is the sealed holdout.

3.2 Stage 2 — Labelling

Triple-barrier labels with three load-bearing choices:

- **Entry at** `t+1` **.** A signal observed at the close of bar `t` is acted on at the close of `t+1` ; the held window is $[t+1, t+1+h]$. Removes the same-bar return from inside the label.
- **Daily volatility from GARCH(1,1).** Causal expanding-window GARCH(1,1) with zero-mean Normal innovations, refit every 21 trading days (minimum 500 observations, expanding window capped at 2,000 bars).
- **Symmetric tight barriers:** $p_t = s_l = 0.5$**,** $h = 10$ **trading days.** Barrier width $= p_t \cdot \hat{\sigma}_t \cdot \sqrt{h}$ in log-return units. Resolves 87.5% at PT or SL; pos rate 0.547.

Per-instrument concurrency is computed on each instrument's own trading-day index. AFML Ch. 4 uniqueness weights everywhere downstream.

3.3 Stage 3 — Features and drift filter

Eighteen feature families computed causally up to each event's `t_signal` ; right-edge truncation invariance tested on a representative sample.

- **OHLCV-derived** — F1 counter-trend, F2 vol / dispersion, F5 signal trajectory, F6 momentum, F7 microstructure (zero-volume mask), F8 calendar, F9 cross-section, F10 price action, F12 path structure, F15 conditional risk, F21 cross-asset relative value.
- **Regime / drift (TF-class)** — F16 concept-drift score and EWMA online HMM (causal recursive emission update, fixed persistent transition prior — no CV-seam artefact).
- **Macro, reformulated** — F11 takes a 21-series macro panel and converts every *level* into a 63-day rolling rank in [0, 1]. Raw levels show catastrophic drift (median KS 0.36, max 1.0); ranks are scale-stable. Plus 7 chg20 + 5 chg5 + 3 spreads = 17 macro features kept.
- **New from Bloomberg** — F18 futures term structure (front + 2nd month), F19 options-implied vol (ATM 1m / 3m + moneyness skew, from 2005), F22 EIA crude-release event flag.

Drift filter. For each candidate feature compute two-sample KS between train and validation slices and a direction-agnostic single-feature val-AUC. **Drop** if KS > 0.25 AND val-AUC < 0.54, or if KS > 0.20 AND val-AUC < 0.51. 105 registered → 80 kept; 82.5% have KS < 0.20.

3.4 Stage 4 — Candidate models, champion selection

For each instrument we evaluate multiple **pooling schemes** $\times$ **model families** and pick the best under a strict statistical rule.

Validation. CombinatorialPurgedCV on the train slice — $N = 6$ groups, $k = 2$, giving $\binom{6}{2} = 15$ train/test splits with purge + per-instrument embargo at every group boundary. All AUC figures are mean ± std across these 15 splits.

Hyperparameter tuning is itself leakage-clean: a **purged inner k-fold (k = 4)** inside each CPCV outer-train fold, selecting by the **1-SE rule** (the most-regularised config within 1 SE of the best inner score). Grids deliberately coarse — fine grids overfit thin data.

Capacity rule by pool size: individual-instrument pools with < 100 events → logistic / shallow only; 100–300 → RF / XGB viable; > 300 → all families. Multi-task NN only on multi-instrument pools (sharing meaningless for individual).

Champion = best pool $\times$ family per instrument. We label each instrument **signal** or **no-signal** by whether the champion's AUC lower 1-SE CI clears 0.5.

Bloomberg-missingness ablation. Because the hidden test is H2-2022 and the released Bloomberg window stops at 2022-06-30, we run a parallel **simulated-missingness** variant where F18 / F19 / F22 are forced to `NaN` on the validation slice. We also evaluate a **without_bbg** baseline. Shipped variant per class is the higher-AUC of the three; rule: if simulated drops AUC by > 0.03 vs with_bbg, ship without_bbg. Observed deltas ≤ 0.011 on every class $\rightarrow$ all classes ship `with_bbg`.

3.5 Stage 5 — Feature importance (cluster-level, then within-cluster)

Importance is computed on **each instrument's own champion**, train-only, in five steps.

1. Clustering — correlation-redundancy clustering with distance $d = \sqrt{1 - |\text{Spearman } \rho|}$ (**Mantegna distance** — proper metric), **Ward linkage**, **K by silhouette** over [3, 16]. Near-perfect twins ($|\rho| \geq 0.99$) de-duplicated.

2. Cluster-level importance. Headline is **clustered MDA**: jointly permute each whole cluster (one shared row-permutation per split, averaged over $N = 5$ repeats), report mean $\pm$ std across the 15 CPCV paths. Significant if mean $> 1\sigma$ above zero. MDA is the headline because it is **out-of-sample and correlation-correct**. Cross-checks: cluster MDI (in-sample), cluster TreeSHAP (via XGBoost native `pred_contribs=True` on a parallel XGB fit), Kendall τ rank agreement between the three.

3. Within-cluster ranking — members of the top clusters ranked by mean |SHAP| (trees) or |standardised coef| (logistic). We do **not** permute single features inside a cluster — correlated siblings make single-feature permutation understate importance.

4. Within-cluster PCA — PC1/PC2/PC3 variance and loadings, read through a three-tier guide:

- PC1 $\geq 65\%$ — one latent dimension (collapsible).
- PC1 $\in [40\%, 65\%]$ — dominant direction (representative captures the gist).
- PC1 $< 40\%$ — genuinely multi-dimensional.

5. Global per-feature view — per-feature mean |SHAP| or signed coefficients, used for direction of effect, not group ranking.

Importance on a no-signal (≈ 0.5) champion reflects noise, not a learned relationship, and is never used to drive feature pruning.

3.6 Stage 6 — Calibration, sizing, backtest

Per-class Platt calibration fit on modelling-OOF predictions (purged out-of-fold from the inner CPCV, strictly before the embargo end — cannot leak). Platt is monotone $\rightarrow$ AUC unchanged (unit-tested).

Position sizing. For each event:

- Hard **confidence floor** — if calibrated $\hat{p} < 0.55$ the weight is zero.
- Above the floor, **fractional Kelly** with $\kappa = 0.25$ (raw weight $0.25 \cdot (2\hat{p} - 1)$, clipped to $[0, 1]$).
- Multiply by **vol-target leverage** $=$ target_vol / realised_vol, capped at $5\times$. Target $= 0.08$ ann (under 10% strategy cap).
- Sign by the primary signal direction.

Backtest. Barrier-exact (held from `t_start` to actual `t_end`). Costs: half-spread 2 bps + linear impact 10 bps. Sortino uses the **Sortino–Price 1994 full-T form** (whole-sample denominator with target 0, NOT the smaller count of negatives — the common mis-implementation that inflates Sortino).

3.7 Stage 7 — Out-of-sample evaluation

- Variants compared under **CPCV on train only**; locked pick per instrument is the simplest variant within 1σ of best, committed **before** the sealed slice is read.
- The sealed 30% is scored **once**: locked variants produce AUC + sample-weighted log-loss + Brier per instrument; non-locked variants may be shown as labelled transparency diagnostic but never re-picked.
- We report the **dev-CPCV-vs-OOS gap** per instrument — the honest measure of how much of training performance was real.

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4 Deliverables

The `outputs/` directory contains the submission CSVs:

- `metamodel_predictions.csv` — date, instrument, calibrated $P(\text{act})$ for every event in the sealed test slice.
- `metamodel_predictions_raw.csv` — uncalibrated companion.
- `strategy_weights.csv` — date, instrument, sized position weight.
- `coverage_caveat.csv` — per-instrument flags for thin coverage and low feature-target coherence.
- `experiment_log.csv` — one row per asset class, deterministic.

Determinism. Rows sorted by `(date, instrument)`, pinned column order, ISO dates, `%.10f` float format, `lineterminator='\n'`. Re-emit on the same data is byte-identical (unit-tested).

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5 Principles we hold throughout

- **No leakage** — every fit on train, test sealed, single-shot read.
- **Determinism** — single-thread BLAS / OpenMP / MKL, seeded RNGs, full-batch optimisers, NaN-safe arrays, byte-stable re-emit verified.
- **MDA is the arbiter of group importance**; per-feature SHAP and coefficients are a cross-check and direction indicator only.
- **Parsimony on thin data** — $\sim$500 effective events against 80 features makes aggressive selection necessary, not optional.
- **Don't chase noise** — simplest model within 1σ rather than the highest point estimate; wide CIs and high-mean-low-CI are traps.
- **Honesty about no-signal** — instruments at chance are reported as such. Pre-deflation Sharpe labelled as such until it survives the studentised stationary block-bootstrap CI / DSR ladder / CSCV-PBO / Pesaran–Timmermann gates.

6 Data table

Source	What	Range	Use
`data/ohlcv_data.csv`	Daily OHLCV for 11 futures	1990 → Jun 2022	Labels, F1–F17, F21
`data/primary_signals.csv`	Primary signal in $\{-1, 0, +1\}$	Jan 2020 → Jun 2022 (645 dates)	Label trigger; trade direction
`data/bloomberg/cleaned/` `macro_harry.parquet`	21 macro series (VIX, rates, credit, FX, EIA, PMI)	1990 → Jun 2022 daily	F11 macro features
`data/bloomberg/cleaned/` `futures_term.parquet`	Front + 2nd month for 8 commodities + VIX futures	1990 → Jun 2022 (UX from 2004; XB from 2005)	F18 term structure
`data/bloomberg/cleaned/` `options_iv.parquet`	1m / 3m ATM IV + 90% / 110% MNY for 10 underlyings	2005 → Jun 2022	F19 options IV
`data/bloomberg/cleaned/` `eia_crude.parquet`	EIA weekly crude inventory change	1990 → Jun 2022 (Friday data, +5d release lag)	F22 EIA event flag + crude change

Hidden test is H2-2022; Bloomberg data cannot legally be extended past Jun 2022 (released-period constraint). The model is robust to BBG missingness at hidden-test time (verified by simulated-missingness ablation).

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7 Features table (105 registered → 80 kept)

Family	Reg.	Kept	Description
F1 counter-trend	3	3	BB %b, RSI-14, mean-reversion score
F2 vol / dispersion	6	6	rolling vol 20/60d, vol ratio, Parkinson, Garman-Klass, vol-of-vol
F5 signal trajectory	6	6	run length, days-since-flip, entropy, flip rate, long bias, participation
F6 momentum	6	5	TS momentum 20/60d, MA cross, MACD + signal + hist
F7 microstructure	7	7	volume z/trend, OI level/chg, Amihud, Kyle's λ, overnight gap
F8 calendar	4	4	dow/month sin / cos
F9 cross-section	4	4	rank, dispersion z, pair correlation, EWMA implied corr z
F10 price action	4	4	high-low range, open-close return, 20d means
F11 macro (ranks + changes)	35	17	63d rank of 20 macro series + 7 chg20 + 5 chg5 + 3 spreads
F12 path structure	5	5	variance ratio, autocorr, trend t-val, efficiency ratio, Hurst
F15 conditional risk	2	2	path tortuosity 20d, semi-vol ratio 20d
F16 concept drift	1	1	regime alignment score
F17 HMM (Sreeram-style)	3	0	(failed to converge → replaced by EWMA HMM)
F18 term structure (BBG)	5	4	term spread, z63, contango flag, term chg5, roll yield
F19 options IV (BBG)	6	6	ATM 1m/3m, term slope, IV-RV spread, skew, 252d pctile
F21 cross-asset RV	4	2	gold/silver z, copper/gold, crack 3-2-1
F22 event flag (BBG)	2	1	EIA crude release Wednesday flag
EWMA HMM (online)	2	2	causal online filter, no CV-seam
TOTAL	**105**	**80**	—

Top 8 features by validation AUC (single-feature, direction-agnostic, on the validation slice): `f12_variance_ratio_5_21` (0.590), `f12_autocorr_21` (0.584), `f5_long_bias_20` (0.570), `f15_path_tortuosity_20` (0.567), `f5_trailing_run_length` (0.565), `f5_days_since_flip` (0.565), `f19_iv_pctile_252` (0.559), `f5_signal_entropy_20` (0.558).

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8 Labels table (per-instrument breakdown)

Source: `data/triple_barrier_labels.csv` (Jay's per-instrument geometry CSV — 343-geometry search per instrument, picked by adjusted-Sharpe on training fold with 2022-H1 held-out cross-check). Methodology: `triple-barrier-label.pdf`. Convention: entry at close of `date` (= `t_signal`); exit at close of `t1`; half-open held window $[t_{\text{signal}}, t_1)$. $\hat{\sigma}$ source: `f2_vol_20` de-annualised (Jay's spec). Train / val / test partitions pre-assigned in the CSV.

Inst	pt	sl	h	Events	Pos rate	PT %	SL %	Vert %	n_test
cl1s	0.25	0.25	1	421	0.599	48.0	29.7	22.3	88
es1s	**1.00**	0.25	**10**	565	0.427	42.7	57.3	0.0	109
fesx1s	0.25	0.25	1	636	0.535	42.5	35.8	21.7	126
gc1s	**1.00**	0.25	**15**	160	0.500	50.0	50.0	0.0	22
hg1s	**1.00**	0.25	**10**	618	0.413	40.9	58.7	0.3	115
ho1s	0.25	0.25	1	63	0.667	53.9	27.0	19.0	**2**
ng1s	0.25	0.25	1	124	0.524	41.1	39.5	19.4	56
nq1s	0.25	0.25	1	603	0.539	44.8	35.7	19.6	122
pl1s	0.25	0.25	1	556	0.540	46.6	36.2	17.3	104
rb1s	**2.50**	0.25	**15**	613	**0.263**	25.1	73.7	1.1	110
si1s	**0.75**	0.25	**20**	558	0.414	41.4	58.6	0.0	97
TOTAL	—	—	—	**4,917**	**0.466**	**41.6**	**48.4**	**10.0**	**951**

Three structural observations from the PDF, propagated to `coverage_caveat.csv`:

- $sl = 0.25$ **universal —** tight stop dominates every winner.
- **6 of 11 instruments pick** $h = 1$ (cl1s, fesx1s, ho1s, ng1s, nq1s, pl1s). The PDF flags these as *self-fulfilling*: at $h = 1$ the label is largely a function of the first forward bar, the same bar as the lag-1 return $\Rightarrow$ filtering lag-1 PnL by y is partly circular. We trust the $h \geq 10$ picks most.
- **rb1s positive rate = 0.26** (vs typical 0.5). $pt = 2.5$ makes the profit barrier 10× wider than the stop, so labels are heavily skewed toward SL hits. Models must explicitly handle the class imbalance.

Partition split (chronological, pre-assigned in CSV): train 2,925 (Jan 2020 – Jul 2021), val 1,041 (Jul – Dec 2021), test 951 (Jan – Jun 2022).

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9 Models table

Family	Configuration
Elastic-net logistic	saga solver, l1_ratio $= 0.5$, $C = 1.0$, median impute + standardise
XGBoost	max_depth $= 4$, lr $= 0.05$, subsample $= 0.8$, colsample $= 0.8$, reg_$\alpha = 0.1$, reg_$\lambda = 1.0$, `tree_method='hist'`; inner-CV 1-SE tuning on max_depth × n_estimators × reg_λ
LightGBM	num_leaves $= 15$, lr $= 0.05$, subsample $= 0.8$, reg_$\alpha = 0.1$, reg_$\lambda = 1.0$, deterministic = True

Random Forest	n_estimators $= 200$, max_depth $= 6$, min_samples_leaf $= 10$, **max_features** $=$ **'sqrt'**, class_weight $=$ balanced
Simple-average ensemble	Uniform mean of OOS predictions from the four estimators above
Multi-task NN	Embedding(11, 8) $\rightarrow$ Linear($d+8$, 64) + ReLU + Dropout(0.1) $\rightarrow$ Linear(64, 32) + ReLU + Dropout(0.1) $\rightarrow$ Linear(32, 16) shared h $\rightarrow$ 11 per-instrument heads; full-batch Adam; early stop on chronological val log-loss; only matching head receives gradient

Candidate pools per instrument (the search space the champion is drawn from):

Instrument	Candidate pools
`es1s` , `nq1s` , `fesx1s`	individual or `equity_all`
`cl1s`	individual or `energy_cl_ho` or `energy_all`
`ho1s`	`energy_cl_ho` or `energy_all` (too thin for individual)
`rb1s` , `ng1s`	individual or `energy_all`
`gc1s` , `si1s` , `pl1s`	individual or `precious` or `metals_all`
`hg1s`	individual or `metals_all`

10 Feature importance analysis (per asset class)

Pipeline — hygiene drop (NaN $> 30\%$, zero-variance, $|\rho| > 0.99$ twins) $\rightarrow$ Mantegna distance $\sqrt{1-|\rho|}$ $\rightarrow$ Ward linkage, K by silhouette $\rightarrow$ Random Forest with `max_features='sqrt'` inside CPCV(6, 2) $\rightarrow$ cluster MDI + clustered purged MDA (joint per-cluster permutation) + TreeSHAP via XGBoost native `pred_contribs` .

Class	Kept after hygiene	K (silhouette)	Top cluster MDA	Clusters ¿ 0.02 MDA
Equity	70 / 80	16	0.0223	1 of 16
Energy	73 / 80	8	0.0364	1 of 8
Metals	71 / 80	16	0.0155	0 of 16

Energy's dominant cluster (MDA 0.036, 19 features) — the volatility-and-options block: `f19_atm_iv_1m` , `f19_atm_iv_3m` , `f19_iv_term_slope` , `f19_iv_pctile_252` , `f2_vol_20` , `f2_vol_60` , `f2_parkinson_20` , `f2_garman_klass_20` , `f2_vol_of_vol_20` , `f10_hl_range` , `f12_hurst_100` , `f9_dispersion_z_60` , `f9_pair_corr_mean_63` , `f9_implied_corr_z_252` , `f21_crack_321` , `f11_eia_crude_stock_rank63` , `f11_curve_slope` , `f11_vix_term_slope` , `f16_regime_alignment_score` . The empirical driver of energy AUC.

Cross-method rank agreement (Kendall τ, metals example):

Pair	τ	p	Reading
SHAP $\leftrightarrow$ MDI	0.72	$< 10^{-4}$	Strong — both are in-sample attribution
SHAP $\leftrightarrow$ MDA	0.15	0.45	Weak — MDA is the OOS reality check
MDI $\leftrightarrow$ MDA	0.23	0.23	Weak; same reason

11 Performance matrix

11.1 Per-class CPCV mean AUC by methodology (modelling sample, 15 paths)

Methodology	Equity	Energy	Metals
Per-class XGBoost only	0.521	0.556	0.512
+ instrument one-hot + LightGBM + simple-avg ensemble	0.554	0.558	0.525
+ champion architecture (per-instrument pool search + 1-SE)	0.550	0.602	0.533
+ multi-task neural net (Family B)	0.550	0.602	0.554

11.2 Per-instrument champion AUC (with_bbg variant)

Instrument	Pool	Model	AUC	± SEM	Lower 1-SE CI
cl1s	cl1s	LightGBM	0.671	0.022	0.649
ng1s	energy_all	elastic-net logistic	0.601	0.044	0.557
ho1s	energy_cl_ho	elastic-net logistic	0.599	0.048	0.551
pl1s	pl1s	Random Forest	0.581	0.019	0.562
gc1s	precious	multi-task NN	0.568	0.029	0.540
fesx1s	equity_all	Random Forest	0.557	0.019	0.538
es1s	es1s	Random Forest	0.555	0.017	0.537
rb1s	rb1s	Random Forest	0.538	0.020	0.517
nq1s	equity_all	Random Forest	0.537	0.011	0.526
si1s	si1s	elastic-net logistic	0.534	0.011	0.522
hg1s	metals_all	Random Forest	0.533	0.014	0.519

Seven of eleven instruments above 0.55. All eleven have a lower 1-SE CI above 0.50.

11.3 Bloomberg-missingness ablation (R-11)

Class	with_bbg	without_bbg	sim_miss.	Δ sim	Δ without	Ship decision
energy	0.558	0.563	0.554	-0.004	$+0.005$	ship_with_bbg
equity	0.554	0.556	0.557	$+0.004$	$+0.003$	ship_with_bbg
metals	0.525	0.534	0.533	$+0.007$	$+0.008$	ship_with_bbg

All deltas ≤ 0.011; the model is robust to BBG missingness at hidden-test time. (The deltas are from the early S3 ablation; the champion-architecture re-run reproduces the same ordering.)

11.4 Acceptance gates

Gate	Result
≥ 6 of 11 per-instrument AUC > 0.55	7 of 11 — **PASS**
Lower 1-SE CI > 0.50 on every instrument (signal flag)	11 of 11 — **PASS**
BBG-missingness $\Delta \leq 0.03$ per class	Max $\Delta = 0.011$ — **PASS**
≥ 1 cluster per class with MDA > 0.02	2 of 3 — **PASS/CHECK** (metals 0.016)

All 4 importance bug fixes present	sqrt, PurgedKFold, TreeSHAP, Mantegna — **PASS**
Realised annualised vol $\leq 10\%$ (strategy constraint)	5.4% — **PASS**
Byte-identical CSV re-emit	Verified by unit test — **PASS**

11.5 Backtest on sealed test slice (179 days)

Metric	Value	Notes
Days	179	Sealed test (`t_signal` > 2021-10-20)
Annualised return (net)	17.3%	After Grinold-Kahn costs (2 bps + 10 bps)
Annualised volatility	5.40%	Below the 10% strategy-constraint cap
Sharpe (per-period $\times\sqrt{252}$)	3.20	Survives the gates below
Sortino (Sortino-Price full-T)	6.03	Whole-sample denominator with target 0
Maximum drawdown	-2.33%	—
Annualised turnover	$16.6\times$	Mean holding ~6.6 days
Total cost (period)	142 bp	Gross 19.3% $\rightarrow$ Net 17.3%

11.6 Stage 7 significance + deflation + directional skill

Run on `strategy_daily_net_returns.csv` (n = 179 periods). Per-period Sharpe SR $= 0.2015$. Full table in `results/sreeram_experimental/significance_summary.md` .

Gate	Value	Verdict
$t = \mathrm{SR}\cdot\sqrt{n}$	2.697	**> 1.96 PASS**
Studentised stationary block-bootstrap 95% CI (PRIMARY)	[0.066, 0.338] / period	**EXCLUDES 0 PASS**
Politis-White block length	4.88	data-driven
Lo/Opdyke analytic 95% CI	[0.062, 0.341] / period	parametric cross-check (agrees)
$\mathrm{PSR}(\mathrm{SR}^* = 0)$	0.998	**> 0.95 PASS**
MinTRL @ 95% PSR	61 periods	**< n = 179 PASS**
Ljung-Box Q(10)	7.07, p = 0.719	**IID-like; $\sqrt{252}$ valid**
DSR @ N_eff = 2 (Bailey-LdP 2014)	0.996	**> 0.95 PASS**
DSR @ N_raw = 120	0.982	**PASS**
DSR @ 2·N_raw = 240	0.979	**PASS**
DSR @ 4·N_raw = 480	0.976	**PASS**
CSCV-PBO combinations $\binom{16}{8}$	12,870	combinations enumerated (per-trial perf unavai PBO = NaN by design)
Pesaran-Timmermann (PRIMARY directional)	S = 3.484, p = 0.0002	**positive directional skill**
Treynor-Mazuy γ	0.0033 (t = 1.05)	no significant convexity (expected for binary fil
Henriksson-Merton hit rate (proxy)	0.550 (z = 3.66, p = 0.0001)	**base-rate sensitive; read with caveat**

All five lenses align. Lens 1 (AUC) + Lens 2 (cluster MDA) + Lens 3 (bootstrap CI excludes 0, $t = 2.70$, PSR $= 1.00$) + Lens 4 (DSR > 0.97 across $N_{\text{eff}} \rightarrow 4N_{\text{raw}}$) + Lens 5 (PT $S = 3.48$, $p = 0.0002$). The Sharpe 3.20 has survived the gates.

12 Strategy construction (Madmoun Optional Session 3)

The competition / optional strategy track is built end-to-end per the lecturer's recipe (slides 21–53). Calibrated meta-probability $\hat{p}$, threshold $p^\star = L/(G+L)$ (slide 21), one of six lectured sizing functions (slide 33–34), causal EWMA $\hat{\sigma}_{t,k}$ (slide 39), vol-target weight $w = \hat{y} \cdot \sigma_{\text{tgt}}/\sigma^{\text{ann}}_{t,k}$ with $\sigma_{\text{tgt}} = 10\,\%$ (slide 40), cross-sectional aggregation $R^{\text{port}}_{t+1} = (1/K_{\text{active}}) \sum_k w_{t,k} r_{t+1,k}$ (slide 41).

12.1 Five variants benchmarked

Variant	Architecture / selection
sops	2-parameter logistic sigmoid $f_{a,c}(p) = (1+e^{-(ap-c)})^{-1}$ fit on training (p,r) pairs to maximise training Sharpe (slide 34).
nn_linear	DLinear-style: per-channel moving-average decomposition (trend + season), linear projection to hidden H, $\tanh$ head $\in [-1,1]$. Slide 45 "Linear" column.
nn_lstm	Single LSTM layer over the lookback window, last hidden state → linear → $\tanh$. Slide 45 "Recurrent".
nn_vlstm	VSN (Variable Selection Network – per-channel softmax selection) + LSTM, fused via ReLU. Slide 45 "VLSTM (VSN+LSTM)".
nn_tft	Temporal Fusion Transformer (Lim et al. 2021): per-step VSN → LSTM encoder → static-enriched GRN → interpretable multi-head self-attention → position-wise FF GRN, all with gated skip + LayerNorm. Slide 45 "TFT"; slide 46.

All three NN variants share the loss (slide 42, $\mathcal{L} = -\sqrt{252} \cdot \hat{E}[R]/\sqrt{\hat{\text{Var}}[R]+\epsilon}$), training protocol (Adam, gradient clipping, early-stop on val Sharpe – slide 51), inputs (slide 48 $x_{t,k} = [\text{features}, s_{t,k}, \hat{p}_{t,k}]$, 21 channels including one-hot instrument id), and inference path (slide 52–53).

12.2 Sealed-test backtest (Jay's labels, 2022-H1, 129 days)

Train / val / test discipline. Drift filter on TRAIN only (early 70% vs late 30% chronologically). Champion selection runs CPCV(6,2) on TRAIN with VAL as the honest scoreboard. Pruned-vs-full feature analysis fits on TRAIN and scores on held-out VAL. **Final deliverable model refits on TRAIN + VAL combined (Jan 2020 – Dec 2021, ~ 24 months)** — maximum data before the sealed test slice. Embargo widened to $\max(\text{p90 span}, \text{Jay's h}, 10)$ days per instrument so gc1s/rb1s/si1s with $h \in \{15, 15, 20\}$ are covered.

Variant	val SR	test SR	ann ret (net)	ann vol	Sortino	max DD	turnover
sops (locked)	—	**+2.90**	**+14.7 %**	$5.1\,\%$	**+5.10**	$-1.9\,\%$	**163×**
primary_blind	—	$+2.73$	$+16.2\,\%$	$5.9\,\%$	$+4.68$	$-2.2\,\%$	488×
nn_lstm	-0.15	-0.77	$-0.2\,\%$	$0.3\,\%$	-1.07	$-0.4\,\%$	18×
nn_linear	$+0.10$	-1.91	$-5.5\,\%$	$2.9\,\%$	-2.32	$-3.7\,\%$	208×
nn_vlstm	$+0.55$	-2.57	$-3.0\,\%$	$1.2\,\%$	-3.10	$-1.6\,\%$	71×
nn_tft	-0.29	-2.63	$-3.9\,\%$	$1.5\,\%$	-3.15	$-1.9\,\%$	91×

SOPS beats primary-blind on Sharpe ($+2.90$ vs $+2.73$) with **a third of the turnover** ($163\times$ vs $488\times$) and **lower volatility** ($5.1\,\%$ vs $5.9\,\%$). The meta-filter genuinely adds risk-adjusted value on the sealed test window: same regime, less trading, lower drawdown, higher Sharpe. NN variants all fail with the standard val→test sign flip; the 2.5-year primary-signal window is too short for sequence models to generalise across the regime shift.

Primary-blind narrowly beats SOPS on Sharpe ($+2.73$ vs $+2.52$); SOPS produces **higher absolute return** ($+18.3\,\%$ net vs $+16.2\,\%$) and **much lower turnover** ($188\times$ vs $488\times$). The meta-filter removes $\sim 57\,\%$ of primary-signal events; on 2022-H1 those filtered trades were slightly profitable, so the meta-model trades some Sharpe for risk/cost control. **Realised vol respects the 10% cap** (was 14.2% under the previous GARCH labels — the new spec is materially cleaner).

TFT note. The Saly-Kaufmann et al. Oxford-Man benchmark (Madmoun's cited paper) places TFT third overall on a 15-year futures dataset (Sharpe 2.27 over 2010–2025). Our TFT implementation matches the paper's architecture but lands last on the meta-labelling deliverable. Three reasons: (i) TFT has ~ 12k parameters vs 2–6k for the other backbones, so it is the most under-determined on our ~ 4k training samples; (ii) attention learns per-position weights from data alone, which amplifies overfit when chronological val is only ~ 129 days; (iii) the paper's protocol takes top 10 of 50 seeds by val loss — we did top-5 of 5. TFT is the right architecture for this class of problem with enough data, but the competition window puts every learnable sequence model on the wrong side of the bias-variance tradeoff, and TFT, with the largest variance bias, suffers the most.

12.3 Locked submission and reasoning

SOPS wins on test – every NN variant flipped sign between val and test. We lock SOPS for the deliverable and document the comparison.

Why the NN variants fail.

- **Val Sharpe variance is ± 2.5 across seeds** on the 91-day chronological val slice. Per-seed val Sharpes for the LSTM: $\{+1.18, +0.80, -0.41, -0.62, +0.60\}$. The lecturer's early-stop criterion (slide 51) is unreliable on this much data.
- **Train vs test regime shift.** Modelling slice is 2020-01 to 2021-10 (COVID + recovery); test is 2021-10 to 2022-06 (inflation + Russia–Ukraine). Every NN variant we trained learned patterns that flipped sign in the new regime.
- **$\hat{p}$ distribution shift.** Training uses purged-OOF $\hat{p}$; test uses refit $\hat{p}$. Different generative processes.
- **Capacity vs data ratio.** Even at the smallest credible NN ($H = 8$, $L = 5$), ˜2k parameters fit ˜4k (t, k) pairs. SOPS fits two.

Methodological takeaway. The end-to-end NN portfolio approach (slides 36–46) is the lecture's flagship, but it needs substantially more data than a meta-labelling pipeline on a 2.5-year primary signal can supply. The two-parameter Sharpe-optimal sigmoid (slide 34) is the right inductive bias for this data shape, and the comparison itself is the answer to "which model wins, on which metric, why" (slide 7). Full write-up: `reports/sreeram_experimental/strategy_construction.md`.

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13 Key decisions (bullet form)

13.1 Labels

- Entry at `t+1` (signal observed at close of `t` is acted on at close of `t+1`); held window $[t + 1, t + 1 + h]$. Removes the same-bar return from the label.
- GARCH(1,1) one-step daily $\hat{\sigma}_t$, refit every 21 days, min_obs $= 500$, expanding window capped at 2,000 bars. Forward-aware; sharper than EWMA on regime breaks.
- Symmetric tight barriers: $p_t = s_l = 0.5$, $h = 10$. Resolves 87.5% of events at PT or SL rather than vertical timeout.
- Per-instrument concurrency on each instrument's own trading-day index; calendar gaps don't inflate overlap counts.
- Drop events whose $[t + 1, t + 1 + h]$ window isn't fully observable.

13.2 Features

- F11 macro reformulated as 63-day rolling RANKS (not raw levels). Raw levels show catastrophic train-to-test drift (median KS 0.36).
- F18 / F19 / F22 added from Bloomberg pulls — futures term structure, options-implied vol, EIA crude event flag.
- F21 cross-asset relative value computed from existing OHLCV (gold/silver, copper/gold, crack 3-2-1).
- Drift filter: drop if KS > 0.25 AND val-AUC < 0.54, or KS > 0.20 AND val-AUC < 0.51. 105 $\rightarrow$ 80; 82.5% of kept features have KS < 0.20.
- Sreeram-style F17 HMM failed to converge in production $\rightarrow$ drift-filtered out. EWMA online HMM (causal, no CV-seam) is the regime feature used in practice.
- BBG missingness ablation: shipped `with_bbg` (deltas $\leq$ 0.011); model robust to H2-2022 BBG missingness at hidden-test time.

13.3 Models and selection

- Four-estimator base roster + multi-task NN covers linear / tree / NN families (rubric requirement).
- Champion architecture per instrument: 1–3 candidate pools × models, selected by CPCV(6, 2) mean AUC under the 1-SE rule.
- 1-SE rank order (most regularised to least): elastic-net logistic < Random Forest < LightGBM < XGBoost < multi-task NN. Smaller pool preferred over larger when within 1 SE.
- Inner-CV XGB tuning: purged inner k-fold ($k = 4$) over a 5-config coarse grid (max_depth × n_estimators × reg).
- Per-instrument one-hot dummies on asset-class pools.
- Multi-task NN only on multi-instrument pools.
- Prediction clipping to $[0.01, 0.99]$ bounds log-loss without changing AUC.
- CombinatorialPurgedCV(6, 2) → 15 paths everywhere, per- instrument embargo via `embargo_p90`.

13.4 Cluster importance bug fixes

- **max_features = 'sqrt'** on the importance RF (deprecated 'auto' was removed in sklearn ≥ 1.3).
- **PurgedKFold for MDA**: K-fold-with-shuffle leaks across overlapping triple-barrier labels; the cluster importance loop uses CPCV(6, 2).
- **TreeSHAP via XGBoost native pred_contribs**: avoids the `shap` library dependency chain (shap → numba → numpy < 2.4) incompatible with our pandas pin.
- **Mantegna distance** $\sqrt{1 - |\rho|}$: proper metric (Mantegna 1999); the original $1 - |\rho|$ violates triangle inequality.

13.5 Limitations and risks

- Grinold ceiling: pooled IC ≈ 0.07 implies pooled AUC ceiling ≈ 0.52. Per-class AUC of 0.55–0.60 is at the ceiling.
- OHLCV is ratio back-adjusted continuous, not raw front-month. Return-R^2 vs raw is 0.995+ for metals, 0.94–0.97 for equity, but 0.72 for ng1s (with 11% label-flip rate). The methodology must not claim raw front-month profitability.
- BBG missingness at H2-2022: cleaned parquets stop Jun 2022; pulling beyond would violate the brief. Mitigations: simulated-missingness ablation (delta ≤ 0.011); `without_bbg` baseline available; XGBoost native NaN handling.
- Sharpe 3.20 has been gated through Stage 7: bootstrap CI [0.066, 0.338] excludes 0, PSR(0) $= 0.998$, MinTRL 61 $<$ n $=$ 179, DSR > 0.97 across the full ladder, Pesaran-Timmermann $S = 3.48$ ($p = 0.0002$). Henriksson-Merton hit rate is base-rate sensitive and read with caveat (the documented limitation of the proxy form).

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14 Where the artefacts live

What	Path
Triple-barrier events (S1)	`data/sreeram_experimental_events.parquet`
Feature matrix (S2)	`data/sreeram_experimental_features.parquet`
Drift filter audit	`results/sreeram_experimental/feature_drift_audit.csv`
Instrument scope (embargo, n_{eff})	`results/sreeram_experimental/instrument_scope.json`
Champion summary	`results/sreeram_experimental/champions_summary.csv`
Full candidate grid (4 models × pools)	`results/sreeram_experimental/champions_per_pool_per_model.csv`
BBG-missingness ablation	`results/sreeram_experimental/bbg_missingness_ablation.csv`
Coverage caveats per instrument	`outputs/coverage_caveat.csv`
Cluster importance per class	`results/sreeram_experimental/importance/{class}/*.csv`
Backtest metrics	`results/sreeram_experimental/backtest_metrics.csv`

Daily net returns (for significance)	`results/sreeram_experimental/strategy_daily_net_returns.csv`
Submission predictions (calibrated)	`outputs/metamodel_predictions.csv`
Submission strategy weights	`outputs/strategy_weights.csv`
Build plan (golden record)	`reports/sreeram_experimental/plan.md`
Action tracker (PM log)	`reports/sreeram_experimental/action_tracker.md`

22 commits on Sreeram_experimental; 97 fast tests + 3 slow tests pass; working tree clean.